STAT2004 – Tutorial Week 8 Solutions

Exercise 1 (Pivot transformation for exponential data)

(a) It is easier to work with the cdf here than the pdf:

$$P(Y \leq y) = P(\lambda X \leq y) = P(X \leq y/\lambda) = F_\lambda(x = y/\lambda) = 1 - \mathrm{e}^{-y},$$

using the cdf of X given in the question. Thus, $Y = \lambda X$ has a standard Exponential(1) distribution which does not depend on any unknown parameters. It is therefore a pivot variable.

(b) There is no unique pair of values a and b satisfying $99\% = P(a \leq Y \leq b)$. One way is to set a as the 0.5%-tile and b as the 99.5%-tile of the Exponential(1) distribution, respectively:

$$\begin{aligned} 0.005 = P(Y \leq a) = 1 - \mathrm{e}^{-a} &\iff a = 0.005 \\ 0.995 = P(Y \leq b) = 1 - \mathrm{e}^{-b} &\iff b = 5.298. \end{aligned}$$

(c) From parts (a) and (b), we have

$$\begin{aligned} 99\% &= P(0.005 \leq Y \leq 5.298) && \text{(cutoffs from part (b))} \\ &= P(0.005 \leq \lambda X \leq 5.298) && \text{(pivot step from part (a))} \\ &= P\left(\frac{0.005}{X} \leq \lambda \leq \frac{5.298}{X}\right). \end{aligned}$$

Thus, a 99% CI for λ can be constructed from the data X via

$$\left(\frac{0.005}{X}, \frac{5.298}{X}\right).$$

(d) For the realisation $x = 12.41882$, the 99% confidence interval is evaluated as

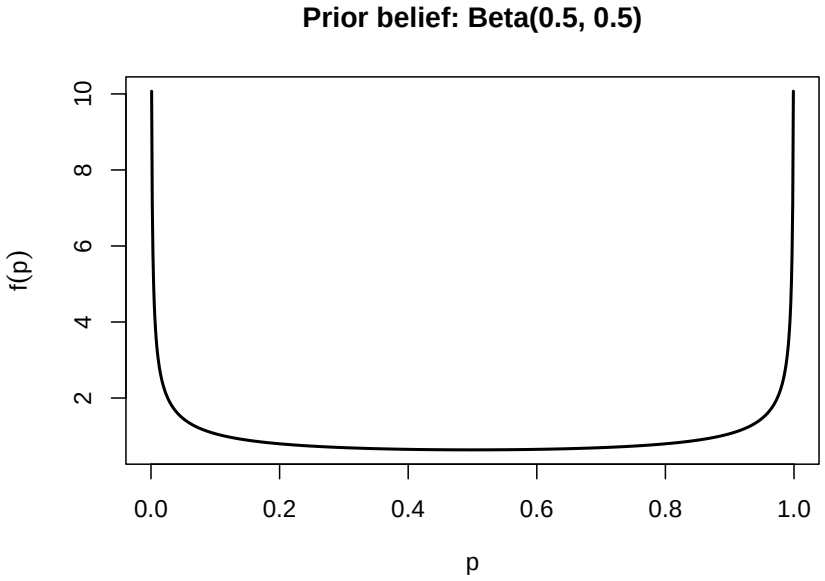
$$\left(\frac{0.005}{12.41882}, \frac{5.298}{12.41882}\right) = (0.0004, 0.427).$$

We can be 99% confident that the true population λ is between $(0.0004, 0.427)$.

(e) We would never know whether our numerical confidence interval $(0.0004, 0.427)$, calculated from one sample of data, contains the true value or not. We only have confidence that the *procedure* in part (c) works in 99% of samples, but there is no guarantee that it worked for this particular sample.

Exercise 2 (Prior and posterior beliefs)

(a) The graph of a Beta(0.5, 0.5) should be U-shaped, with peaks at $p = 0$ and $p = 1$, and a valley around $p = 0.5$.




```
p <- seq(0.001, 0.999, length.out = 1000)
prior <- dbeta(p, 0.5, 0.5)

plot(p, prior, type = "l", lwd = 2,
     xlab = expression(p), ylab = expression(f(p)),
     main = "Prior␣belief:␣Beta(0.5,␣0.5)")
```

(b) This prior belief is qualitatively the opposite of Max's belief, because we are now placing very little belief around $p = 0.5$, and instead placing most of our belief around 0 and 1. In other words, our prior belief is that the true Skittles proportion is highly skewed one way or another, but it's *not* going to be around 50%.

(c) By Bayes' theorem, the posterior is proportional to the likelihood times the prior:

$$f(p \mid x = 23) \propto f(x = 23 \mid p)\, f(p).$$

Since $X \mid p \sim \text{Binomial}(n = 47, p)$, the likelihood is

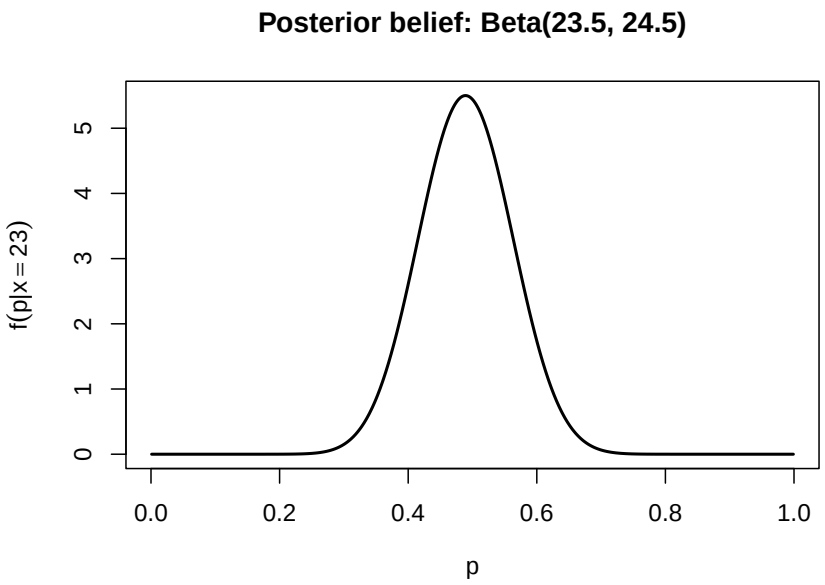
$$f(x = 23 \mid p) = \binom{47}{23} p^{23}(1-p)^{47-23} \propto p^{23}(1-p)^{24}.$$

Multiplying the prior in (a) with the likelihood of our observed data $p^{23}(1-p)^{24}$ gives the desired result:

$$f(p \mid x = 23) \propto p^{23}(1-p)^{24} \cdot p^{0.5-1}(1-p)^{0.5-1} = p^{23+0.5-1}(1-p)^{24+0.5-1}.$$

This posterior is a Beta(23.5, 24.5) distribution, which has a mountain shape with a hump at $23.5/(23.5 + 24.5)$.

(d)




```
p <- seq(0.001, 0.999, length.out = 1000)
posterior <- dbeta(p, 23.5, 24.5)

plot(p, posterior, type = "l", lwd = 2,
     xlab = expression(p), ylab = expression(f(p * "|" * x == 23)),
     main = "Posterior␣belief:␣Beta(23.5,␣24.5)")
```

(e) Even though our starting belief is that the Skittle population might be highly skewed either way (closer to 0% or closer to 100%), after observing the dataset of $x = 23$ Skittles out of $n = 47$ samples, our belief has now shifted away from the extremes. The bulk of our belief is now closer to the middle in the parameter space where the observed data suggests.